

Equivariant Jiang–Su Stability for Amenable Actions in the Unital Stably Finite Case

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Abstract

Every action of a countable discrete amenable group on a simple, separable, unital, infinite-dimensional, nuclear, stably finite complex C^* -algebra that is already Jiang–Su stable absorbs the trivial action on the Jiang–Su algebra up to cocycle conjugacy. No restriction is imposed on the action on the tracial-state simplex. This proves the unital, stably finite case of Szabó’s Conjecture A on automatic equivariant Jiang–Su stability.

1 Introduction

Jiang–Su stability is a regularity property of a C^* -algebra that also has a dynamical counterpart: an action should absorb the trivial action on the Jiang–Su algebra up to cocycle conjugacy. The theorem below establishes this implication for all amenable actions in the simple, nuclear, finite unital setting. Throughout, tensor products are minimal, and $\mathcal{Z}$ denotes the Jiang–Su algebra.

Theorem 1.1. *Let A be a simple, separable, unital, infinite-dimensional, nuclear, stably finite complex C^* -algebra satisfying $A \cong A \otimes \mathcal{Z}$. Let G be a countable discrete amenable group and let $\alpha : G \rightarrow \text{Aut}(A)$ be any action. Put*

$$B = A \otimes \mathcal{Z}, \quad \beta_g = \alpha_g \otimes \text{id}_{\mathcal{Z}}.$$

There are a unital $$ -isomorphism $\Phi : A \rightarrow B$ and unitaries $u_g \in B$, $g \in G$, such that*

$$u_e = 1, \quad u_{gh} = u_g \beta_g(u_h), \quad (1.1)$$

$$\Phi(\alpha_g(a)) = u_g \beta_g(\Phi(a)) u_g^* \quad (g \in G, a \in A). \quad (1.2)$$

Here stable finiteness means that every isometry in every matrix algebra over A is unitary. Amenability is used in its Følner form: for every finite $H \subset G$ and $\delta > 0$, some nonempty finite $F \subset G$ satisfies $|hF \triangle F| < \delta|F|$ for $h \in H$. Theorem 1.1 imposes no outerness, Rokhlin, or UCT assumption, and no condition on the extreme tracial boundary or on trace orbits. In particular, it includes finite groups and trivial actions.

1.1 Context and scope

Szabó formulated automatic equivariant Jiang–Su stability for amenable actions on separable simple nuclear Jiang–Su-stable algebras as Conjecture A of [2]. Theorem 1.1 resolves its unital, stably finite case. The original formulation also includes nonunital algebras, which lie beyond the scope of the present theorem. Earlier affirmative cases include finite algebras with finitely many rays of extremal traces and very weak comparison [2, Theorem C].

Szabó and Wouters proved that equivariant uniform property Gamma characterizes equivariant Jiang–Su stability in the setting used here [3, Corollary C]. Their theorem supplies the absorption step of the proof. Our task is to construct the required invariant central trace splittings for an arbitrary action. These splittings are uniform over the entire trace space, even when the action moves traces.

1.2 Construction and proof structure

The technical argument has three parts. First, tensorial absorption supplies countably many commuting *orbit algebras* inside the uniform tracial central sequence algebra. Under every extreme trace on the algebra they generate, distinct orbit algebras are independent. This intrinsic independence is useful because the traces under consideration need not extend to the ambient algebra. We then prove a joint central limit statement for normalized sums and their empirical covariances, allowing the extreme trace to vary with the number of rows.

Second, the covariance is incorporated into a quadratic Gaussian construction. For a Følner set F , its Gram vectors define a positive operator P_F of trace one. Taking a polynomial approximation to $P_F^{1/2}$ gives a quadratic form whose variance stays bounded away from zero and whose change under translation is small. This square-root normalization handles covariance matrices of arbitrary rank, including highly correlated ones. Empirical covariances turn the same formula into exactly equivariant noncommutative polynomials. Compactness of the finite covariance space then promotes the scalar moment limits to estimates uniform over every tracial state.

Finally, weighting traces by elements of A yields a central invariant unitary with Haar distribution relative to A . Two explicit diagonal selections give this unitary and then equal trace projections. The construction of the rows appears in Section 3, the uniform unitary estimates in Section 4, and the diagonal selections and absorption argument in Section 5. All parts of this construction are proved below; the final absorption theorem is stated separately in Theorem 2.2.

Several ingredients can be used separately under their stated hypotheses: intrinsic independence for commuting rows (Lemma 3.2), moment estimates under varying tracial states (Proposition 3.3), and convergence of scalar spectral laws against bounded continuous functions without uniform operator-norm bounds (Lemma 4.4).

2 Traces and the absorption criterion

For a unital C^* -algebra D , write $T(D)$ for its tracial-state space. For $\tau \in T(D)$, set

$$\|x\|_{2,\tau} = \tau(x^*x)^{1/2}.$$

Under the assumptions of Theorem 1.1, $T(A)$ is a nonempty compact convex set. Indeed, nuclearity implies exactness, and a unital stably finite exact algebra has a tracial state; moreover its 2-quasitraces are traces [1, Theorem 5.11 and Corollary 5.12]. Every tracial state on the simple algebra A is faithful: its null space $\{a : \tau(a^*a) = 0\}$ is a closed two-sided ideal that does not contain 1.

Fix a free ultrafilter ω on $\mathbb{N} = \{1, 2, \dots\}$. Define

$$\|a\|_{2,u} = \sup_{\tau \in T(A)} \|a\|_{2,\tau}, \quad A^\omega = \ell^\infty(\mathbb{N}, A)/I_\omega, \quad I_\omega = \{(a_j) : \lim_{j \rightarrow \omega} \|a_j\|_{2,u} = 0\}. \quad (2.1)$$

The inequalities $\|ab\|_{2,u} \leq \|a\| \|b\|_{2,u}$ and $\|ba\|_{2,u} \leq \|a\| \|b\|_{2,u}$ show that I_ω is a closed two-sided ideal. Faithfulness of traces embeds A as the constant sequences in A^ω . For any

sequence (τ_j) in $T(A)$, the formula

$$\tau([(a_j)]) = \lim_{j \rightarrow \omega} \tau_j(a_j)$$

defines a tracial state on A^ω , since $|\tau_j(a_j)| \leq \|a_j\|_{2,\tau_j}$. These are the *limit traces*, denoted $T_\omega(A)$. We do not identify them with all tracial states on A^ω .

Automorphisms permute $T(A)$ and therefore preserve $\|\cdot\|_{2,u}$. Applying α_g to representative sequences defines an action $\bar{\alpha} : G \curvearrowright A^\omega$. It preserves $A^\omega \cap A'$. The quotient in Equation (2.1) is also the quotient of the norm ultrapower by its uniform trace-kernel ideal, since norm-null sequences belong to I_ω .

Definition 2.1. The action α has *equivariant uniform property Gamma* if, for every integer $k \geq 2$, there are pairwise orthogonal projections $p_1, \dots, p_k \in (A^\omega \cap A')^{\bar{\alpha}}$ with

$$\tau(ap_i) = \frac{\tau(a)}{k} \quad (a \in A, \tau \in T_\omega(A), 1 \leq i \leq k). \quad (2.2)$$

This is the unital simple formulation in [3, Definition A]. Our construction also gives $\sum_i p_i = 1$. We use the following established theorem.

Theorem 2.2 (Szabó–Wouters). *Let a countable discrete amenable group act on a separable, unital, simple, nuclear, $\mathcal{Z}$ -stable C^* -algebra with nonempty tracial-state space. If the action has equivariant uniform property Gamma, it is cocycle conjugate to its tensor product with the trivial action on $\mathcal{Z}$.*

Theorem 2.2 is the implication of [3, Corollary C] needed here. Thus it remains to construct the projections required by Definition 2.1. The next two sections work inside a separable subalgebra C of $A^\omega \cap A'$. There we use *all* traces in $T(C)$; this will allow weighted restrictions of limit traces in Section 5.

3 Commuting orbit rows and joint moments

Throughout this section, A , G , and α satisfy the hypotheses of Theorem 1.1. Only countability of G is needed for the constructions in this section.

Proposition 3.1 (Commuting orbit rows). *There are unital embeddings $\iota_i : \mathcal{Z} \rightarrow A^\omega$, $i \geq 1$, for which the unital orbit algebras*

$$C_i = C^*(\bar{\alpha}_g(\iota_i(\mathcal{Z})) : g \in G)$$

commute with A and commute pairwise. In particular,

$$C = C^*\left(\bigcup_{i \geq 1} C_i\right) \subseteq A^\omega \cap A'$$

is unital, separable, and $\bar{\alpha}$ -invariant, and $T(C) \neq \emptyset$.

Moreover, there are self-adjoint elements $x_{i,g} \in C_i$, $i \geq 1$, $g \in G$, and a constant $M < \infty$, such that

$$\|x_{i,g}\| \leq M, \quad \varphi(x_{i,g}) = 0, \quad \varphi(x_{i,g}^2) = 1 \quad (\varphi \in T(C)), \quad (3.1)$$

and

$$\bar{\alpha}_h(x_{i,g}) = x_{i,hg} \quad (i \geq 1, g, h \in G). \quad (3.2)$$

Proof. The Jiang–Su algebra is simple, unital, uniquely tracial, and isomorphic to its infinite tensor power; see [4, Example 1.14(iv) and Corollary 1.11]. Write ρ for its unique tracial state. Since A is $\mathcal{Z}$ -stable, fix an identification

$$A \cong A \otimes \bigotimes_{l=1}^{\infty} \mathcal{Z}.$$

We use the increasing finite tensor stages in this identification.

We first show that every norm-separable C^* -subalgebra $D \subseteq A^\omega$ admits a unital embedding

$$\mathcal{Z} \longrightarrow A^\omega \cap D'. \quad (3.3)$$

Choose a norm-dense sequence $(d_s)_{s \geq 1}$ in D , and choose bounded representative sequences $d_s = [(d_{s,j})_{j \geq 1}]$. For each j , choose a finite tensor stage containing elements within $1/j$ in norm of $d_{1,j}, \dots, d_{j,j}$. Let $\theta_j: \mathcal{Z} \rightarrow A$ be the embedding into a tensor factor strictly after this stage. For $j \geq s$ and $z \in \mathcal{Z}$,

$$\|[\theta_j(z), d_{s,j}]\| \leq \frac{2\|z\|}{j}.$$

The coordinatewise homomorphisms therefore induce a unital $*$ -homomorphism $z \mapsto [(\theta_j(z))_j]$ into $A^\omega \cap D'$. It is injective because $\mathcal{Z}$ is simple and the unit of A^ω is nonzero. This proves Equation (3.3).

Set $D_0 = A$. Inductively, suppose that D_{i-1} is a separable, unital, $\bar{\alpha}$ -invariant subalgebra of A^ω containing A and the preceding orbit algebras. Apply Equation (3.3) to obtain $\iota_i(\mathcal{Z}) \subseteq D'_{i-1}$, define C_i as in the statement, and put

$$D_i = C^*(D_{i-1}, C_i).$$

For $b \in \iota_i(\mathcal{Z})$, $d \in D_{i-1}$, and $g \in G$,

$$[\bar{\alpha}_g(b), d] = \bar{\alpha}_g([b, \bar{\alpha}_{g^{-1}}(d)]) = 0,$$

since D_{i-1} is invariant. Thus C_i commutes with D_{i-1} . Countability of G preserves separability at each stage. This gives the required algebras and their commutation properties. A limit trace on A^ω restricts to a tracial state on C , so $T(C)$ is nonempty.

The trace ρ is faithful by simplicity of $\mathcal{Z}$. Choose a non-scalar self-adjoint element $c \in \mathcal{Z}$. Then $\rho((c - \rho(c)1)^2) > 0$, so rescaling $c - \rho(c)1$ gives a self-adjoint $b \in \mathcal{Z}$ with $\rho(b) = 0$ and $\rho(b^2) = 1$. Set

$$x_{i,g} = \bar{\alpha}_g(\iota_i(b)), \quad M = \|b\|.$$

Every $\varphi \in T(C)$ restricts along each translated unital embedding of $\mathcal{Z}$ to ρ . This proves Equation (3.1), and Equation (3.2) follows from the action law. $\square$

The following independence statement is intrinsic to the algebra generated by the rows. In particular, it applies to traces of C that do not extend to the ambient ultrapower.

Lemma 3.2 (Independence at extreme traces). *Let B be a unital C^* -algebra generated by pairwise commuting unital C^* -subalgebras B_i , $i \geq 1$. If $\varphi \in \partial_e T(B)$, then*

$$\varphi(b_1 \cdots b_s) = \prod_{j=1}^s \varphi(b_j)$$

whenever $b_j \in B_{i_j}$ and the indices $i_1, \dots, i_s$ are distinct.

Proof. Let (H_φ, π, Ω) be the tracial GNS representation of φ , and put

$$\mathcal{M} = \pi(B)'' , \quad \mathcal{M}_i = \pi(B_i)'' .$$

We first record the relevant properties of this representation, allowing φ to have a nonzero kernel on B . On the dense subspace of vectors $[a]$, $a \in B$, right multiplication $R_b[a] = [ab]$ is bounded with norm at most $\|b\|$, by traciality. It commutes with $\pi(B)$, while the vectors $R_b\Omega = [b]$ are dense. If $T \in \mathcal{M}$ satisfies $T\Omega = 0$, then

$$TR_b\Omega = R_bT\Omega = 0 \quad (b \in B),$$

so $T = 0$. Thus Ω is separating for $\mathcal{M}$. The normal vector state $\tau(T) = \langle T\Omega, \Omega \rangle$ is consequently faithful: for $T \geq 0$, $\tau(T) = 0$ implies $T^{1/2}\Omega = 0$. Its traciality extends from $\pi(B)$ to $\mathcal{M}$ by separate ultraweak continuity. Hence $\mathcal{M}$ is a finite von Neumann algebra with faithful normal tracial state τ .

Extremality of φ implies that $\mathcal{M}$ is a factor. Indeed, a nontrivial central projection $p \in \mathcal{M}$ would give the convex decomposition

$$\varphi(a) = \tau(p) \frac{\tau(p\pi(a))}{\tau(p)} + \tau(1-p) \frac{\tau((1-p)\pi(a))}{\tau(1-p)} .$$

The two normalized functionals are tracial states on B . They are distinct: if, for instance, the first equalled φ , equality of the corresponding normal functionals would extend from $\pi(B)$ to $\mathcal{M}$, and evaluation at p would give $1 = \tau(p)$. This contradicts $0 < \tau(p) < 1$.

The algebras $\mathcal{M}_i$ commute and generate $\mathcal{M}$. Every element of the center of $\mathcal{M}_i$ therefore commutes with all of $\mathcal{M}$, so

$$Z(\mathcal{M}_i) \subseteq Z(\mathcal{M}) = \mathbb{C}1 .$$

The restriction of τ makes each $\mathcal{M}_i$ a finite factor with a faithful normal tracial state.

Fix i , and let $y \geq 0$ belong to the von Neumann algebra generated by the other $\mathcal{M}_j$'s. It commutes with $\mathcal{M}_i$. The functional

$$\psi_y(x) = \tau(y^{1/2}xy^{1/2}) = \tau(xy), \quad x \in \mathcal{M}_i,$$

is normal and positive: sandwiching by $y^{1/2}$ preserves suprema of bounded increasing nets of positive operators. Commutation of y with $\mathcal{M}_i$ makes it tracial, and $0 \leq \psi_y \leq \|y\|\tau$. The Radon–Nikodym Theorem for normal functionals on a finite von Neumann algebra therefore gives a density $h \in \mathcal{M}_i$, $0 \leq h \leq \|y\|1$, with $\psi_y(x) = \tau(hx)$. Invariance of ψ_y under unitary conjugation forces h to commute with every unitary in $\mathcal{M}_i$. Hence h is central and thus scalar. Evaluating at the unit identifies that scalar as $\tau(y)$. It follows that

$$\tau(xy) = \tau(x)\tau(y) .$$

Linearity extends this identity to arbitrary y in the algebra generated by the other rows. Applying it successively proves the required product formula. $\square$

We now fix the algebras and elements from Proposition 3.1. For $N \geq 1$ and $g, h \in G$, set

$$z_{N,g} = \frac{1}{\sqrt{N}} \sum_{i=1}^N x_{i,g}, \quad k_{N;g,h} = \frac{1}{N} \sum_{i=1}^N \frac{x_{i,g}x_{i,h} + x_{i,h}x_{i,g}}{2} . \quad (3.4)$$

The last two subscripts of $k_{N;g,h}$ are covariance indices. These elements are self-adjoint, $k_{N;g,h} = k_{N;h,g}$, and

$$\bar{\alpha}_l(z_{N,g}) = z_{N,lg}, \quad \bar{\alpha}_l(k_{N;g,h}) = k_{N;lg,lh} \quad (l, g, h \in G). \quad (3.5)$$

Proposition 3.3 (Joint ordered moment limit). *Let $J \subseteq G$ be nonempty and finite. Let $N_j \rightarrow \infty$ be a sequence of positive integers, and let $\varphi_j \in \partial_e T(C)$. Suppose that*

$$\varphi_j(k_{N_j;g,h}) \longrightarrow R_{g,h} \quad (g, h \in J). \quad (3.6)$$

Then $R = (R_{g,h})_{g,h \in J}$ is a real positive semidefinite matrix with diagonal entries one. Choose unit vectors v_g , $g \in J$, spanning a finite-dimensional real Euclidean space L , with $\langle v_g, v_h \rangle = R_{g,h}$. If W is a standard Gaussian vector in L and $\xi_g = \langle v_g, W \rangle$, then

$$((z_{N_j;g})_{g \in J}, (k_{N_j;g,h})_{g,h \in J}) \longrightarrow ((\xi_g)_{g \in J}, (R_{g,h})_{g,h \in J})$$

in all ordered joint moments under φ_j . Equivalently, for every noncommutative $$ -polynomial P in these self-adjoint variables,*

$$\varphi_j(P((z_{N_j;g})_g, (k_{N_j;g,h})_{g,h})) \longrightarrow \mathbb{E} P((\xi_g)_g, (R_{g,h})_{g,h}). \quad (3.7)$$

For every sequence (N_j, φ_j) as above, a subsequence satisfying Equation (3.6) exists.

Proof. For a tracial state φ and self-adjoint x, y ,

$$\varphi(xy) = \varphi(yx) = \overline{\varphi(xy)}.$$

Thus all the second moments under consideration are real and symmetric. For real scalars a_g ,

$$\sum_{g,h \in J} a_g a_h \varphi_j(k_{N_j;g,h}) = \frac{1}{N_j} \sum_{i=1}^{N_j} \varphi_j \left(\left(\sum_{g \in J} a_g x_{i,g} \right)^2 \right) \geq 0.$$

The diagonal entries are one by Equation (3.1). Cauchy–Schwarz bounds every entry in absolute value by one. Consequently these covariance matrices lie in a compact set of real symmetric matrices; their limits have the asserted properties, and the final subsequence assertion follows. A Gram representation of the positive semidefinite matrix R supplies L and the vectors v_g .

The empty word is immediate. We first consider a word of length $d \geq 1$ in the fluctuation variables. At one term of the given sequence write $N = N_j$ and $\varphi = \varphi_j$. Expanding the sums gives

$$\varphi(z_{N,g_1} \cdots z_{N,g_d}) = N^{-d/2} \sum_{i_1, \dots, i_d=1}^N \varphi(x_{i_1,g_1} \cdots x_{i_d,g_d}). \quad (3.8)$$

Partition the positions $1, \dots, d$ according to equality of their row labels i_a . Since distinct rows commute, we may group the factors belonging to each row, preserving their order within that row. Lemma 3.2 then factors the expectation over the distinct rows. A singleton block gives zero by Equation (3.1). Each remaining summand has absolute value at most M^d .

A partition with b blocks has at most N^b choices of distinct row labels. Its contribution to Equation (3.8) is therefore bounded by $M^d N^{b-d/2}$. Among partitions without singleton blocks, $b \leq \lfloor d/2 \rfloor$. It follows that odd moments tend to zero, and, when $d = 2q$, only partitions into q pairs can have a nonzero limit.

Fix such a pairing, writing its pairs as (a_l, b_l) , $1 \leq l \leq q$, with $a_l < b_l$. Its contribution is

$$N^{-q} \sum_{\substack{1 \leq i_1, \dots, i_q \leq N \\ i_1, \dots, i_q \text{ distinct}}} \prod_{l=1}^q \varphi(x_{i_l, g_{a_l}} x_{i_l, g_{b_l}}).$$

Dropping the distinctness condition changes this by $O_{d,M}(N^{-1})$: there are at most $\binom{q}{2}N^{q-1}$ tuples with a collision, and each product is bounded by M^{2q} . The unrestricted sum factors, and traciality identifies its factors with the empirical covariances:

$$\prod_{l=1}^q \left(\frac{1}{N} \sum_{i=1}^N \varphi(x_{i,g_{a_l}} x_{i,g_{b_l}}) \right) = \prod_{l=1}^q \varphi(k_{N;g_{a_l},g_{b_l}}).$$

All error estimates depend only on d and M , so they hold uniformly for the moving traces φ_j . By Equation (3.6), the limiting even moment is the sum over pairings of the corresponding products of entries of R . This is the Gaussian moment: the moment-generating function of $(\xi_g)_{g \in J}$ is

$$\mathbb{E} \exp \left(\sum_{g \in J} t_g \xi_g \right) = \exp \left(\frac{1}{2} \sum_{g,h \in J} t_g t_h R_{g,h} \right),$$

whose Taylor expansion gives precisely this pairing formula.

It remains to include the covariance variables in arbitrary positions of a word. For $g, h \in J$, define

$$s_{i;g,h} = \frac{x_{i,g}x_{i,h} + x_{i,h}x_{i,g}}{2}, \quad y_{j,i;g,h} = s_{i;g,h} - \varphi_j(s_{i;g,h})1.$$

This is centering separately in each row, at the current trace. In particular,

$$\varphi_j(y_{j,i;g,h}) = 0, \quad \|y_{j,i;g,h}\| \leq 2M^2,$$

and there is the exact identity

$$k_{N_j;g,h} - \varphi_j(k_{N_j;g,h})1 = \frac{1}{N_j} \sum_{i=1}^{N_j} y_{j,i;g,h}. \quad (3.9)$$

Let U_j be an ordered word having d fluctuation factors and $r \geq 1$ factors of the form on the left of Equation (3.9). Expand all its sums. Each of its $d+r$ row factors is centered under φ_j , and a singleton row again gives zero by Lemma 3.2. A surviving term therefore uses at most $\lfloor (d+r)/2 \rfloor$ distinct row labels. The products within any one row have absolute expectation bounded by the product of the operator norms of their factors. There are only finitely many partitions of the $d+r$ positions. Together with the normalization $N_j^{-d/2-r}$, these observations give

$$|\varphi_j(U_j)| \leq C_{d,r,M} N_j^{\lfloor (d+r)/2 \rfloor - d/2 - r} \leq C_{d,r,M} N_j^{-r/2} \longrightarrow 0, \quad (3.10)$$

where the constant is independent of j and φ_j .

Finally, replace each covariance factor in a general word by its scalar φ_j -mean plus the centered factor in Equation (3.9). Every term containing a centered covariance factor tends to zero by Equation (3.10). The remaining terms converge by the fluctuation calculation and Equation (3.6). This proves convergence of every ordered word. Since all generators are self-adjoint, linearity also gives the $*$ -polynomial formulation in Equation (3.7). $\square$

4 Gaussian quadratics over Følner sets

We retain the algebra C , its invariant commuting orbit algebras C_i , and the variables $x_{i,g}$, $z_{N,g}$, and $k_{N;g,h}$ from Section 3. In particular, $T(C)$ is nonempty, the $x_{i,g}$ are uniformly bounded by M , and Proposition 3.3 applies to arbitrary sequences of extreme traces. All traces of finite-dimensional matrices in this section are unnormalized. For a matrix B , we write $\|B\|_{\text{HS}} = (\text{Tr}(B^*B))^{1/2}$ and $\|B\|_1 = \text{Tr} |B|$.

Proposition 4.1 (Uniform approximately invariant unitaries). *For every finite set $H \subset G$, every $\varepsilon > 0$, and every integer $m \geq 1$, there are $t > 0$ and a self-adjoint element $Q \in C$ such that $V = \exp(itQ)$ satisfies*

$$\sup_{\varphi \in T(C)} \|\bar{\alpha}_h(V) - V\|_{2,\varphi} < \varepsilon \quad (h \in H), \quad (4.1)$$

$$\sup_{\varphi \in T(C)} |\varphi(V^\ell)| < \varepsilon \quad (1 \leq \ell \leq m). \quad (4.2)$$

We first record the matrix and probability estimates used in the proof.

Lemma 4.2 (Square-root perturbation bound). *If S, T are positive matrices of the same size, then*

$$\|S^{1/2} - T^{1/2}\|_{\text{HS}}^2 \leq \|S - T\|_1.$$

Proof. Set $a = S^{1/2}$, $b = T^{1/2}$, $d = a - b$, and $u = \text{sign}(d)$, with $\text{sign}(0) = 0$. Since $a^2 - b^2 = ad + db$, cyclicity of the trace gives

$$\text{Tr}(u(a^2 - b^2)) = \text{Tr}(|d|(a + b)).$$

In a basis diagonalizing d , positivity gives $a_{jj} + b_{jj} \geq |a_{jj} - b_{jj}| = |d_j|$. Thus

$$\text{Tr}(|d|(a + b)) = \sum_j |d_j|(a_{jj} + b_{jj}) \geq \sum_j d_j^2 = \text{Tr}(d^2).$$

On the other hand, u is a contraction, so trace-norm duality gives

$$\text{Tr}(u(a^2 - b^2)) \leq \text{Tr}|a^2 - b^2|.$$

These two inequalities prove the assertion. $\square$

Lemma 4.3 (Gaussian quadratic estimates). *Let W be a standard Gaussian vector in a finite-dimensional real Euclidean space, and let B be a real symmetric operator on that space. Put*

$$q_B = \langle W, BW \rangle - \text{Tr}(B).$$

Then

$$\|q_B\|_{L^2} = \sqrt{2}\|B\|_{\text{HS}}, \quad |\mathbb{E} \exp(isq_B)| \leq (1 + 4s^2\|B\|_{\text{HS}}^2)^{-1/4} \quad (s \in \mathbb{R}). \quad (4.3)$$

Its law has an exponential moment of its absolute value and is determined by its moments.

Proof. Orthogonal diagonalization of B gives

$$q_B \stackrel{\text{law}}{=} \sum_{j=1}^d \lambda_j (U_j^2 - 1),$$

where λ_j are its eigenvalues and U_j are independent standard real normal variables. The summands are centered and have variances $2\lambda_j^2$, proving the first formula. The one-dimensional Gaussian integral gives

$$|\mathbb{E} \exp(isq_B)| = \prod_{j=1}^d (1 + 4s^2\lambda_j^2)^{-1/4}.$$

For nonnegative numbers a_j , one has $\prod_j (1 + a_j) \geq 1 + \sum_j a_j$. Applying this inequality with $a_j = 4s^2\lambda_j^2$ proves the second formula in Equation (4.3).

Choose $c > 0$ so that $2c \max_j |\lambda_j| < 1$; if $B = 0$, any $c > 0$ will do. The pointwise estimate

$$\exp\left(c \left| \sum_j \lambda_j (U_j^2 - 1) \right| \right) \leq \exp\left(c \sum_j |\lambda_j| \right) \prod_j \exp(c |\lambda_j| U_j^2)$$

and the Gaussian integral show that $\mathbb{E} \exp(c |q_B|) < \infty$.

For completeness, let ν be another probability measure with the same moments as q_B . By the nonnegative power series for cosh and monotone convergence,

$$\int_{\mathbb{R}} \cosh(cx) d\nu(x) = \sum_{r=0}^{\infty} \frac{c^{2r}}{(2r)!} \mathbb{E}(q_B^{2r}) = \mathbb{E} \cosh(cq_B) < \infty.$$

Since $e^{c|x|} \leq 2 \cosh(cx)$, the measure ν also has an exponential moment. The characteristic functions of ν and of q_B extend analytically to the strip $\{z \in \mathbb{C} : |\operatorname{Im} z| < c\}$. Their Taylor coefficients at zero agree, so the Identity Theorem makes these analytic functions equal. Uniqueness of characteristic functions proves that ν is the law of q_B . $\square$

The purpose of square-root normalization is already visible for a positive real symmetric matrix P with $\operatorname{Tr}(P) = 1$. Equation (4.3) gives

$$\operatorname{Var}(q_{P^{1/2}}) = 2, \quad \operatorname{Var}(q_P) = 2 \operatorname{Tr}(P^2).$$

The latter variance is 2 when P is a rank-one projection, but $2/n$ when $P = n^{-1}I_n$. The exact square root therefore prevents the variance from vanishing as the rank grows. Lemma 4.2 simultaneously controls its change under perturbation. The proof below replaces this exact square root by a polynomial with an explicitly controlled error.

Lemma 4.4 (From moments to bounded continuous functions). *Let $X_j = X_j^*$ belong to unital C^* -algebras equipped with states φ_j . Let q_B be a centered Gaussian quadratic as in Lemma 4.3. Suppose that*

$$\varphi_j(X_j^d) \longrightarrow \mathbb{E}(q_B^d) \quad (d = 0, 1, 2, \dots).$$

Then, for every bounded continuous function $f : \mathbb{R} \rightarrow \mathbb{C}$,

$$\varphi_j(f(X_j)) \longrightarrow \mathbb{E}f(q_B).$$

No uniform bound on $\|X_j\|$ is needed.

Proof. Let μ_j be the spectral probability law of X_j under φ_j . The second moments of μ_j are bounded, so the sequence is tight. Fix an integer $d \geq 0$, and choose an even integer $2r > d$. Moment convergence supplies a constant $C_r < \infty$ such that $\int |x|^{2r} d\mu_j(x) \leq C_r$ for all j . Consequently

$$\int_{|x| > K} |x|^d d\mu_j(x) \leq C_r K^{d-2r} \quad (K > 0). \quad (4.4)$$

Consider any weakly convergent subsequence, with limit μ . Lower semicontinuity gives $\int |x|^{2r} d\mu(x) \leq C_r$, so the same tail bound holds for μ . Multiply x^d by a continuous cutoff which equals one on $[-K, K]$ and zero outside $[-2K, 2K]$. Weak convergence applies to this bounded continuous function. Letting K tend to infinity and using Equation (4.4) shows that

$$\int x^d d\mu(x) = \mathbb{E}(q_B^d).$$

This holds for every d . Lemma 4.3 therefore identifies μ with the law of q_B . Tightness now shows that the whole sequence μ_j converges weakly to that law: every subsequence has a weakly convergent further subsequence, and all such limits have just been identified. This is precisely the claimed convergence for f . $\square$

Proof of Proposition 4.1. Step 1. Choice of parameters. Fix H, ε, m as in the statement. Choose $t > 0$ so that

$$(1 + t^2)^{-1/4} < \varepsilon/4. \quad (4.5)$$

Next choose $\delta > 0$ and $0 < \eta < 1/2$ so that

$$t\sqrt{2}(\sqrt{\delta} + 2\eta) < \varepsilon/4. \quad (4.6)$$

Amenability gives a nonempty finite $F \subset G$ satisfying

$$|hF \triangle F| < \delta|F| \quad (h \in H).$$

Set

$$n = |F|, \quad J = F \cup \bigcup_{h \in H} hF.$$

Finally, choose a real polynomial

$$p(s) = \sum_{r=1}^D c_r s^r, \quad \sup_{0 \leq s \leq 1} |p(s) - \sqrt{s}| < \eta/\sqrt{n}. \quad (4.7)$$

Such a polynomial exists: first approximate $\sqrt{s}$ on $[0, 1]$ by a real polynomial to within $\eta/(2\sqrt{n})$, then subtract its value at zero. Thus t, δ, η, F, n, J , and p , including its degree and coefficients, are all fixed before N tends to infinity. None is chosen using a trace or a limiting covariance.

Step 2. Uniform Gaussian comparison. Let $R = (R_{g,g'})_{g,g' \in J}$ be any real positive semidefinite matrix with diagonal one. Choose unit vectors v_g with Gram matrix R , spanning a finite-dimensional real Euclidean space L , and let W be standard Gaussian in L . Write $\xi_g = \langle v_g, W \rangle$. For $E = F$ or $E = hF$, define

$$P_E = \frac{1}{n} \sum_{g \in E} v_g v_g^\top, \quad B_E = p(P_E), \quad q_E = \langle W, B_E W \rangle - \text{Tr}(B_E).$$

The operator P_E is positive with trace one, so its spectrum lies in $[0, 1]$; its rank is at most n . Because $p(0) = 0$, the operator $p(P_E) - P_E^{1/2}$ vanishes on the kernel of P_E . The scalar approximation in Equation (4.7) therefore gives

$$\|B_E - P_E^{1/2}\|_{\text{HS}}^2 \leq \text{rank}(P_E) \eta^2/n \leq \eta^2, \quad \|P_E^{1/2}\|_{\text{HS}}^2 = \text{Tr}(P_E) = 1. \quad (4.8)$$

In particular,

$$1 - \eta \leq \|B_E\|_{\text{HS}} \leq 1 + \eta. \quad (4.9)$$

The summands indexed by $F \cap hF$ cancel when subtracting P_F and P_{hF} . Each remaining $v_g v_g^\top$ has trace norm one. Hence Lemma 4.2 implies

$$\|P_F^{1/2} - P_{hF}^{1/2}\|_{\text{HS}}^2 \leq \|P_F - P_{hF}\|_1 \leq \frac{|F \triangle hF|}{n} < \delta.$$

Combining this with Equation (4.8) yields

$$\|B_F - B_{hF}\|_{\text{HS}} \leq \sqrt{\delta} + 2\eta.$$

Both forms q_F, q_{hF} are evaluated on the same W , so their difference is the centered quadratic associated with $B_{hF} - B_F$. Lemma 4.3 gives

$$\|q_{hF} - q_F\|_{L^2} \leq \sqrt{2}(\sqrt{\delta} + 2\eta). \quad (4.10)$$

Also, Equation (4.9) and $\eta < 1/2$ give, for every integer $\ell \geq 1$,

$$|\mathbb{E} \exp(i\ell t q_F)| \leq (1 + 4\ell^2 t^2 \|B_F\|_{\text{HS}}^2)^{-1/4} \leq (1 + t^2)^{-1/4}. \quad (4.11)$$

All these bounds hold for every such R , including singular matrices. No invariance of R under translation of its indices has been used.

Step 3. Equivariant polynomial prescription. For $r \geq 1$, direct multiplication of the rank-one summands gives

$$P_E^r = \frac{1}{n^r} \sum_{g_1, \dots, g_r \in E} R_{g_1, g_2} \cdots R_{g_{r-1}, g_r} v_{g_1} v_{g_r}^\top.$$

Taking its quadratic form and its trace, respectively, shows that

$$q_E = \sum_{r=1}^D \frac{c_r}{n^r} \sum_{g_1, \dots, g_r \in E} R_{g_1, g_2} \cdots R_{g_{r-1}, g_r} (\xi_{g_1} \xi_{g_r} - R_{g_r, g_1}). \quad (4.12)$$

The product of covariance entries is interpreted as 1 when $r = 1$.

Use the same prescription in C by setting

$$\tilde{Q}_{N,E} = \sum_{r=1}^D \frac{c_r}{n^r} \sum_{g_1, \dots, g_r \in E} k_{N;g_1, g_2} \cdots k_{N;g_{r-1}, g_r} (z_{N, g_1} z_{N, g_r} - k_{N;g_r, g_1}), \quad (4.13)$$

$$Q_{N,E} = \frac{1}{2} (\tilde{Q}_{N,E} + \tilde{Q}_{N,E}^*). \quad (4.14)$$

Multiplication in Equation (4.13) is in the displayed order. The entries of the empirical covariance array need satisfy no matrix positivity condition: this is a finite polynomial in elements of C , whereas square-root functional calculus was applied only to P_E in the Gaussian comparison.

The exact relabeling in Equation (3.5) carries each tuple in the sum for F to its left translate in the sum for hF . It preserves the order of all factors, and taking self-adjoint parts commutes with $\bar{\alpha}_h$. Consequently

$$Q_{N, hF} = \bar{\alpha}_h(Q_{N, F}) \quad (h \in H, N \geq 1). \quad (4.15)$$

Step 4. Transfer of moments and scalar laws. Consider now any sequence $N_j \rightarrow \infty$ and extreme traces $\varphi_j \in \mathcal{T}(C)$ such that

$$\varphi_j(k_{N_j; g, g'}) \longrightarrow R_{g, g'} \quad (g, g' \in J).$$

Proposition 3.3 gives convergence of every ordered mixed moment of the empirical variables. Any moment of the polynomials in Equations (4.13)–(4.14) is a finite linear combination of such moments. Since the limiting variables commute and Equation (4.12) is real-valued, it follows that the $Q_{N_j, E}$ converge jointly in moments under φ_j to the q_E . In particular,

$$\|Q_{N_j, hF} - Q_{N_j, F}\|_{2, \varphi_j} \longrightarrow \|q_{hF} - q_F\|_{L^2}. \quad (4.16)$$

Lemma 4.4 also gives

$$\varphi_j(\exp(isQ_{N_j, F})) \longrightarrow \mathbb{E} \exp(isq_F) \quad (s \in \mathbb{R}). \quad (4.17)$$

This use of scalar spectral laws is pertinent because the operator norms of the $Q_{N,E}$ may grow with N . Indeed, the elementary estimates $\|k_{N; g, g'}\| \leq M^2$ and $\|z_{N, g}\| \leq \sqrt{N} M$ yield only

$$\|Q_{N,E}\| \leq (N+1) \sum_{r=1}^D |c_r| M^{2r}.$$

Convergence of the scalar moments, together with the tail estimate Equation (4.4), is sufficient for Equation (4.17).

Step 5. Uniformity over all traces. Set $V_N = \exp(itQ_{N,F})$. For any self-adjoint X, Y in a unital C^* -algebra and any tracial state φ , differentiation gives

$$\frac{d}{ds}(e^{i(1-s)tX}e^{istY}) = e^{i(1-s)tX}it(Y - X)e^{istY}.$$

The tracial 2-seminorm is unchanged by left and right unitary multiplication. Integration and the triangle inequality therefore give

$$\|e^{itX} - e^{itY}\|_{2,\varphi} \leq t\|X - Y\|_{2,\varphi}. \quad (4.18)$$

In view of Equations (4.15), (4.16), and (4.10), every sequence of traces and indices with convergent covariance as above satisfies

$$\limsup_{j \rightarrow \infty} \|\bar{\alpha}_h(V_{N_j}) - V_{N_j}\|_{2,\varphi_j} \leq t\sqrt{2}(\sqrt{\delta} + 2\eta) < \varepsilon/4, \quad (4.19)$$

$$\lim_{j \rightarrow \infty} |\varphi_j(V_{N_j}^\ell)| = |\mathbb{E} \exp(i\ell t q_F)| \leq (1 + t^2)^{-1/4} < \varepsilon/4. \quad (4.20)$$

Here $h \in H$ and $1 \leq \ell \leq m$, and the strict inequalities are Equations (4.6) and (4.5).

It follows that, for all sufficiently large N , every extreme trace $\varphi \in T(C)$ satisfies

$$\|\bar{\alpha}_h(V_N) - V_N\|_{2,\varphi} < \varepsilon/2 \quad (h \in H), \quad |\varphi(V_N^\ell)| < \varepsilon/2 \quad (1 \leq \ell \leq m). \quad (4.21)$$

Indeed, otherwise choose $N_j \rightarrow \infty$ and an extreme trace φ_j for which one of these finitely many tests fails. Passing to a subsequence fixes the failed test. The matrices

$$(\varphi_j(k_{N_j;g,g'}))_{g,g' \in J}$$

belong to the compact set of real positive semidefinite matrices with diagonal one. A further subsequence has a limiting covariance R , and Equation (4.19) or Equation (4.20) contradicts the failure at tolerance $\varepsilon/2$. Only the finite covariance matrices were extracted; no compactness of the extreme trace boundary is needed. Nor is a convergence rate uniform in R required: each putative sequence of failures supplies one limiting R , to which the preceding moment argument applies. Moment determinacy is used after that R has been fixed, so no common exponential-moment radius over all covariance matrices is required.

Fix an N for which Equation (4.21) holds. For each $h \in H$, the squared seminorm

$$\varphi \mapsto \varphi((\bar{\alpha}_h(V_N) - V_N)^*(\bar{\alpha}_h(V_N) - V_N))$$

is weak-star continuous and affine on $T(C)$. For each $1 \leq \ell \leq m$, the map $\varphi \mapsto |\varphi(V_N^\ell)|$ is weak-star continuous and convex. The sublevel sets with bounds $\varepsilon^2/4$ and $\varepsilon/2$, respectively, are closed and convex and contain all extreme traces. By the Krein–Milman Theorem they contain $T(C)$. Thus the suprema in Equations (4.1) and (4.2) are at most $\varepsilon/2$, and in particular strictly less than ε . Taking $Q = Q_{N,F}$ and $V = V_N$ completes the proof. $\square$

5 Invariant trace splittings and absorption

We pass from Proposition 4.1 to exact relations in A^ω . We first record the coordinate-selection fact used in both passages.

Lemma 5.1 (Coordinate selection). *If $x = [(x_j)] \in A^\omega$ has a bounded representative, then*

$$\sup_{\tau \in T_\omega(A)} \|x\|_{2,\tau} = \lim_{j \rightarrow \omega} \sup_{\sigma \in T(A)} \|x_j\|_{2,\sigma}, \quad (5.1)$$

$$\sup_{\tau \in T_\omega(A)} |\tau(x)| = \lim_{j \rightarrow \omega} \sup_{\sigma \in T(A)} |\sigma(x_j)|. \quad (5.2)$$

Consequently, finitely many such tests can be realized simultaneously at a single coordinate with any prescribed positive slack above their ultralimit bounds.

Proof. Every sequence of traces gives the upper bound in each identity. Conversely, for each j , choose $\sigma_j \in T(A)$ within $1/j$ of the supremum for the test in question. Its limit trace realizes the reverse inequality; square roots and absolute values commute with ultralimits of bounded scalar sequences. For finitely many tests, the coordinates satisfying each individual bound with positive slack form a set in ω . Their intersection is again in ω , hence nonempty and indeed contains arbitrarily large coordinates. The maximizing trace may be different for every test. $\square$

5.1 A weighted Haar unitary

Keep the invariant separable algebra $C \subset A^\omega \cap A'$ from Proposition 3.1. For $a \in A_+$ and $\tau \in T_\omega(A)$, define on C

$$\tau_a(c) = \tau(ac). \quad (5.3)$$

This is a positive tracial functional. Positivity follows from $ac = a^{1/2}ca^{1/2}$ when $c \geq 0$, since a commutes with C . For $c, d \in C$, traciality and the same commutation give $\tau(acd) = \tau(dac) = \tau(adc)$. Its norm is $\tau(a)$. If $\tau(a) > 0$, its normalization is therefore an element of $T(C)$; if $\tau(a) = 0$, the functional is zero. Proposition 4.1 thus gives, for its unitary V ,

$$|\tau(aV^\ell)| < \varepsilon \tau(a) \leq \varepsilon \quad (0 \leq a \leq 1, \tau \in T_\omega(A), 1 \leq \ell \leq m, \tau(a) > 0), \quad (5.4)$$

with zero left side when $\tau(a) = 0$. The approximate invariance estimates apply to limit traces by restriction to C .

Proposition 5.2 (A weighted Haar unitary). *There is a unitary $w \in (A^\omega \cap A')^{\bar{\alpha}}$ such that, for every $a \in A$, $\tau \in T_\omega(A)$, and $f \in C(\mathbb{T})$,*

$$\tau(af(w)) = \tau(a) \int_{\mathbb{T}} f d\mu, \quad (5.5)$$

where $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$ and μ is normalized Haar measure.

Proof. Choose increasing finite sets $H_m \subset G$ whose union is G , and a sequence $a_1, a_2, \dots$ dense in the positive unit ball of A , with $a_1 = 1$. Apply Proposition 4.1 with $(H, \varepsilon, m) = (H_m, 1/m, m)$, and denote its exponential unitary by $V^{(m)} \in C$. It has a representative $(v_j^{(m)})_j$ consisting of unitaries in A : lift its self-adjoint exponent to a bounded self-adjoint sequence and exponentiate coordinatewise. The bounds for these self-adjoint lifts may depend on m , whereas the unitary representatives always have norm one.

Since $V^{(m)}$ commutes with A , Lemma 5.1, Equation (5.4), and the invariance estimates allow a coordinate $j(m) \geq m$ at which $w_m = v_{j(m)}^{(m)}$ satisfies

$$\begin{aligned} \| [w_m, a_r] \|_{2,u} &< 2/m & (1 \leq r \leq m), \\ \| \alpha_h(w_m) - w_m \|_{2,u} &< 2/m & (h \in H_m), \\ \sup_{\sigma \in T(A)} |\sigma(a_r w_m^\ell)| &< 2/m & (1 \leq r, \ell \leq m). \end{aligned} \quad (5.6)$$

For the last line, apply Equation (5.2) to the class $a_r(V^{(m)})^\ell$. All coordinate tests are finite in number, and their suprema already range over all traces.

The class $w = [(w_m)]$ is a unitary. The first two lines of Equation (5.6), density, and $\|[w_m, a - a_r]\|_{2,u} \leq 2\|a - a_r\|$ show that it is central relative to A and fixed by $\bar{\alpha}$. The last line, first for every a_r , then by norm density for positive contractions, and then by linearity for $a \in A$, gives

$$\tau(aw^\ell) = 0 \quad (a \in A, \tau \in T_\omega(A), \ell \geq 1).$$

The same holds for negative powers, because $\tau(aw^{-\ell}) = \overline{\tau(a^*w^\ell)}$. The zeroth power has trace $\tau(a)$. Thus Equation (5.5) holds for trigonometric polynomials. Uniform approximation on $\mathbb{T}$ and $|\tau(af(w))| \leq \|a\|\|f\|_\infty$ prove it for every continuous f . $\square$

5.2 Equal trace projections

Proposition 5.3 (Equal trace projections). *The action α has equivariant uniform property Gamma. The projections in Definition 2.1 can be chosen to sum to 1.*

Proof. Fix $k \geq 2$ and let $w = [(w_j)]$ be the unitary of Proposition 5.2, represented by unitaries $w_j \in A$. Partition $\mathbb{T}$ into k half-open arcs $I_1, \dots, I_k$ of Haar measure $1/k$, and write $\chi_i = 1_{I_i}$. Choose continuous functions $f_i^{(r)} : \mathbb{T} \rightarrow [0, 1]$ satisfying

$$e_r := \max_{1 \leq i \leq k} \|f_i^{(r)} - \chi_i\|_{L^2(\mu)} \longrightarrow 0.$$

For example, replace each indicator near its two endpoints by linear ramps, with the total lengths of the transition arcs tending to zero. For this proof fix r , and write $f_i = f_i^{(r)}$ and $e = e_r$. Pointwise bounds and the triangle inequality imply

$$\begin{aligned} \|f_i^2 - f_i\|_{L^2(\mu)} &\leq e, \\ \|f_i f_j\|_{L^2(\mu)} &\leq 2e \quad (i \neq j), \\ \left\| \sum_i f_i - 1 \right\|_{L^2(\mu)} &\leq ke, \\ \left| \int_{\mathbb{T}} f_i d\mu - \frac{1}{k} \right| &\leq e. \end{aligned} \tag{5.7}$$

For the first inequality, use $|f_i^2 - f_i| \leq |f_i - \chi_i|$ almost everywhere. For the second, expand $f_i f_j - \chi_i \chi_j = (f_i - \chi_i)f_j + \chi_i(f_j - \chi_j)$ and use $\chi_i \chi_j = 0$ almost everywhere. The remaining inequalities follow from $\sum_i \chi_i = 1$ almost everywhere and the L^1 – L^2 inequality for a probability measure.

Set $y_i^{(r)} = f_i^{(r)}(w)$. These are central, invariant positive contractions, since continuous functional calculus respects commutation and automorphisms. Equation (5.5) with $a = 1$, applied to the squared modulus of each continuous function in Equation (5.7), gives the corresponding 2-norm bounds for $y_i^{(r)}$, uniformly over $\tau \in T_\omega(A)$. For every positive contraction $a \in A$, it also gives

$$\left| \tau(ay_i^{(r)}) - \frac{\tau(a)}{k} \right| = \tau(a) \left| \int_{\mathbb{T}} f_i^{(r)} d\mu - \frac{1}{k} \right| \leq e_r. \tag{5.8}$$

Represent $y_i^{(r)}$ by $f_i^{(r)}(w_j)$. Use H_r and $a_1, \dots, a_r$ from the proof of Proposition 5.2. For this fixed r , Lemma 5.1 gives a coordinate $j(r) \geq r$ such that the positive contractions

$y_{i,r} = f_i^{(r)}(w_{j(r)})$ satisfy, simultaneously for all indicated indices,

$$\begin{aligned}
& \|y_{i,r}^2 - y_{i,r}\|_{2,u} < e_r + 1/r, \\
& \|y_{i,r}y_{j,r}\|_{2,u} < 2e_r + 1/r \quad (i \neq j), \\
& \left\| \sum_i y_{i,r} - 1 \right\|_{2,u} < ke_r + 1/r, \\
& \|[y_{i,r}, a_s]\|_{2,u} < 1/r \quad (s \leq r), \\
& \|\alpha_h(y_{i,r}) - y_{i,r}\|_{2,u} < 1/r \quad (h \in H_r), \\
& \sup_{\sigma \in T(A)} \left| \sigma(a_s y_{i,r}) - \frac{\sigma(a_s)}{k} \right| < e_r + 1/r \quad (s \leq r).
\end{aligned} \tag{5.9}$$

This is a coordinate choice after $f_i^{(r)}$ has been fixed. It therefore needs no uniform continuity modulus, as r varies, for these increasingly sharp arc approximations.

Let $p_i = [(y_{i,r})_r] \in A^\omega$. Their representatives are uniformly bounded positive contractions. The first three lines of Equation (5.9) imply that p_i are orthogonal projections summing to 1; the next two lines imply centrality and invariance by density. For every sequence (σ_r) in $T(A)$, the last line gives

$$\lim_{r \rightarrow \omega} \sigma_r(a_s y_{i,r}) = \frac{1}{k} \lim_{r \rightarrow \omega} \sigma_r(a_s).$$

This is Equation (2.2) for a_s , for every limit trace. Norm density and linearity extend it to all $a \in A$. Thus the projections satisfy Definition 2.1. $\square$

5.3 The cocycle equations

Proof of Theorem 1.1. The trace hypotheses were checked in Section 2, and Proposition 5.3 supplies equivariant uniform property Gamma. All assumptions of Theorem 2.2 hold. It follows that α and $\beta = \alpha \otimes \text{id}_{\mathcal{Z}}$ are cocycle conjugate. We spell out the orientation to obtain the exact equations of the theorem.

An isomorphism $\Phi : A \rightarrow B$ is automatically unital, since it is surjective between unital algebras. Put $\gamma_g = \Phi \alpha_g \Phi^{-1}$. If cocycle conjugacy is supplied in the form $\gamma_g = \text{Ad}(u_g) \beta_g$ for a β -cocycle u , then Equations (1.1) and (1.2) follow directly. In the reverse orientation, suppose

$$\beta_g = \text{Ad}(v_g) \gamma_g, \quad v_{gh} = v_g \gamma_g(v_h).$$

Here the unitaries lie in B , whose multiplier algebra is B itself. Set $u_g = v_g^*$. Then

$$u_g \beta_g(u_h) = v_g^* v_g \gamma_g(v_h^*) v_g^* = \gamma_g(v_h^*) v_g^* = v_{gh}^* = u_{gh}, \quad \text{Ad}(u_g) \beta_g = \gamma_g.$$

Thus either orientation gives the required β -cocycle. Its identity relation $u_e = u_e^2$, together with unitarity, forces $u_e = 1$. This proves both displayed equations and hence the theorem. $\square$

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